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Lightweight Multi-Agent System for Questionnaire-based Lung Cancer Early Screening

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Abstract

The proportion of late-stage lung cancer diagnoses has remained persistently high. Although low-dose spiral computed tomography (LDCT) is effective, its widespread adoption at community level remains challenging due to constraints such as cost, resource availability and radiation exposure. In order to address this issue, the project proposes MAQ-SCREEN: a questionnaire-driven multi-agent lung cancer early screening system tailored for community and primary healthcare settings. The system operates on a multi-agent collaborative architecture, incorporating functional agents for question generation, conversational interaction, answer verification, data analysis, and report generation. Through message coordination and semantic reasoning, it enables dynamic question-answering, real-time validation, and interpretable risk assessment. The system has been designed to target individuals aged 45 and above, and it integrates Automatic Speech Recognition (ASR) and Text-to-Speech (TTS) in order to lower digital barriers. The cross-platform web solution is characterised by its ability to function without the need for installation. In order to ensure the security and protection of personal data, a multifaceted approach is employed, incorporating encryption, access controls, and anonymisation mechanisms. These mechanisms are implemented at all stages of the data lifecycle, including collection, transmission, and storage. The system also provides user data management and follow-up comparison capabilities. The MAQ-SCREEN system, a questionnaire-based lightweight Multi-Agent Lung Cancer Screening System, outputs personalised risk grades, tailored recommendations, and visualised reports. The tool under discussion serves the function of a pre-screening triage instrument for LDCT. The purpose of this instrument is threefold: firstly, to enhance early detection rates, secondly, to optimise resource allocation, and thirdly, to provide anonymised data support for epidemiological and public health policy. The project is congruent with SDG 3 - Good Health and Well-being, with the objective being the establishment of a cost-effective, interpretable, and scalable pathway for lung cancer screening that does not rely on costly imaging equipment.

1. Introduction

Lung cancer is among the most prevalent and lethal malignant tumours in the world [1]. As demonstrated in the statistical data published by the World Health Organisation in 2024, the annual global mortality rate from lung cancer surpasses the aggregate of deaths from colorectal, breast, and prostate cancers combined [2]. In Malaysia, lung cancer is

similarly a significant public health concern. As stated in the Malaysian Cancer Registry’s summary of grade reports, lung cancer was the third most prevalent cancer in the country from 2017 to 2021 [3]. It is noteworthy that approximately 92%–95% of lung cancer cases are diagnosed at stage III or IV. While the five-year survival rate for patients with stage I lung cancer is 37.1%, this figure is significantly lower, at 6.3%, for stage IV patients [4]. The findings of this study unequivocally demonstrate that early screening and diagnosis are paramount for enhancing patient survival rates.

In accordance with the national clinical practice guidelines for lung cancer, as published in 2025 by organisations including the Malaysian Lung Cancer Network, the primary early screening method in the country involves the administration of LDCT scans to high-risk populations. The objective of this approach is to enhance early detection rates and reduce the proportion of late-stage diagnoses [5]. Although LDCT is widely recognised as the most effective lung cancer screening method currently available, its high cost, relatively high false-positive rate, radiation exposure risks, and significant demand on healthcare resources have, to some extent, limited its wider adoption and implementation. Consequently, guidelines generally recommend focusing LDCT screening primarily on high-risk populations. This, however, gives rise to a critical question: namely, how to identify potential high-risk individuals suitable for LDCT screening in a more economical and efficient manner. In order to address this challenge, the project has designed a questionnaire-based, lightweight Multi-Agent Screening System for Lung Cancer (MAQ-SCREEN). The overarching objective of this system is to facilitate low-cost lung cancer risk screening, thus compensating for the resource and economic limitations inherent to conventional LDCT screening methodologies.

1.1 Project Background

From the perspective of global research and technological development, intelligent agent studies for early-stage lung cancer screening remain in their infancy. However, in other disease domains, systems employing artificial intelligence for early intelligent screening and auxiliary diagnosis have achieved significant progress, providing crucial reference points for the design of this project. Three representative solutions—Cognoa’s Canvas Dx, Qingdao University–Huawei’s Aiwon Pancreatic Cancer Early Screening research, and Eyenuk EyeArt’s Diabetic Retinopathy Screening System—demonstrate distinct technical approaches and application models in multimodal data fusion, deep learning modelling, and clinical implementation.

1.2 Existing Solutions / Similar Applications

Existing Solutions / Similar Applications

Description:
 ✓✓ indicates a significant advantage;
 ✓ indicates the feature is present;
 ✗ indicates the feature is not present.

	MAQ-SCREEN	Cognoa Canvas Dx	Aiwen Pancreatic System	Eyenuk EyeArt
AI-Driven Screening Capability	✓	✓	✓	✓
Multimodal Data Integration	✓	✓	✓	✗
Operational Independence	✓✓	✗	✗	✓
Accessibility in Primary Healthcare / Community	✓✓	✗	✗	✓
Cost-Effectiveness and Portability	✓✓	✗	✗	✗

Figure 1: Comparison of Existing AI-Based Screening Systems

Cognoa Canvas Dx represents a significant development in the field of autism spectrum disorder diagnosis, as it is the first system to receive approval from the US Food and Drug Administration for use in the diagnosis of ASD using artificial intelligence [6]. The system integrates three types of input data—parental questionnaires, child behavioural videos, and clinician observations—and employs gradient-boosted decision tree models for multimodal fusion analysis. In clinical trials, the system demonstrated high accuracy, with a positive predictive value of 80.8% and a negative predictive value of 98.3% [7]. This approach has been demonstrated to significantly reduce the diagnostic cycle time and enhance the diagnostic capabilities of primary care providers. However, due to its reliance on multiple inputs, the presence of some “uncertain” results, and its inability to diagnose independently, it remains solely an adjunct to clinical decision-making [?]. The Qingdao University–Huawei pancreatic cancer early screening research employs a stratified screening strategy, commencing with low-cost preliminary screening via questionnaires and basic health data, followed by in-depth CT imaging analysis for high-risk individuals. Its core model is two-stage and weakly supervised: nnU-Net first localises the

pancreas, then a multiple-instance learning stage separates tumour from normal tissue, achieving high accuracy and sensitivity on real-world data [8]. Eyenuk EyeArt represents a significant innovation as the world’s first FDA-approved fully autonomous diabetic retinopathy screening system capable of operating independently within primary healthcare settings; clinical studies report sensitivity of 96% and specificity of 88%, and deployment at scale [9].

2. Problem Statements

Lung cancer is among the most lethal malignant tumours globally, consistently occupying the foremost position among cancer-related causes of death. Notwithstanding the continuous advancements in medical imaging and clinical treatment, the preponderance of lung cancer cases remains diagnosed at an advanced stage. According to data from the Malaysian Cancer Registry, approximately 92%–95% of lung cancer patients are already at stage III or IV at the time of diagnosis, at which point the curative potential of treatment is significantly diminished. The five-year survival rate for patients diagnosed at an early stage is approximately 37.1%, whereas for patients diagnosed at an advanced stage it plummets to just 6.3% [4]. These figures underscore two key points. Firstly, they highlight the critical importance of early detection. Secondly, they demonstrate the persistent gap in reality.

LDCT is recognised on an international level as the most effective method for the screening of lung cancer in its early stages. Nevertheless, the expense, the potential for radiation exposure, and the need for substantial healthcare resources pose significant obstacles to its large-scale implementation in the general population. For economically underdeveloped regions or primary healthcare facilities, the capital expenditure and maintenance costs associated with LDCT equipment constitute the primary barriers to universal adoption [10]. Consequently, a significant number of individuals with a high probability of being at high risk are unable to access screening services in a timely manner. This results in delayed detection of early-stage lung cancer patients.

In Malaysia, the inequitable distribution of healthcare resources serves to compound this issue [11]. While urban areas tend to have relatively comprehensive imaging facilities, remote communities and primary care clinics frequently lack the necessary diagnostic capabilities. In circumstances where low-cost screening tools are unavailable, a considerable number of unmet needs remain with regard to the early detection of lung cancer.

In response to this situation, the project proposes MAQ-SCREEN, a questionnaire-based,

lightweight multi-agent lung cancer early screening system. The system employs a multi-agent collaborative mechanism to conduct semantic analysis, logical reasoning, and risk assessment on user questionnaire information. This enables it to automatically identify high-risk individuals and generate personalised screening recommendations. The objective of this solution is to provide an economical, efficient, and scalable alternative pathway for lung cancer early screening in settings where resources are limited.

3. Project Objectives

- **Questionnaire-driven module:** personalised question generation, response validation, comprehensive input analysis, risk calculation, and visualised report output.
- **Voice interaction module:** ASR/TTS for users aged 45+ to lower digital barriers and improve accessibility and usability.
- **User data management module:** secure storage of screening history (questionnaires, risk scores, recommendations) with longitudinal comparison and trend analysis.
- **Privacy protection module:** encryption, access control, anonymisation, logging, and backup to prevent unauthorised access and data loss across the data lifecycle.

4. Motivation

The motivation for developing MAQ-SCREEN arises from the persistently high rate of late-stage lung cancer diagnoses in Malaysia and worldwide, highlighting the urgent need for accessible early screening. Despite advances in diagnostic imaging, limited availability of affordable and scalable early screening tools in community healthcare remains a major obstacle. Integrating intelligent technologies into the screening process is expected to enhance efficiency and reduce lung cancer mortality.

This project aims to improve health risk awareness among middle-aged and elderly populations, who often neglect screening due to limited knowledge or access to information. By introducing interactive voice questionnaires, users can conveniently perform self-assessments and receive personalised health guidance.

Furthermore, current lung cancer screening information is fragmented and difficult for the public to interpret. MAQ-SCREEN addresses this by establishing a unified, trust-

worthy digital platform that integrates validated screening criteria and interpretable risk assessment methods. It also explores multi-agent collaboration to automate question generation, answer validation, and report creation—improving accuracy, efficiency, and resource allocation for LDCT screening. Ultimately, MAQ-SCREEN seeks to provide an accessible, comprehensible, and sustainable intelligent solution for early lung cancer screening.

5. Proposed Solution

In light of the aforementioned problem analysis, the objective of this project is to develop a low-cost, intelligent early-stage lung cancer screening system that can be applied in community and primary healthcare settings. In this regard, the MAQ-SCREEN methodology is proposed as a means of achieving non-invasive, personalised risk assessment. The system’s design objectives align with the United Nations Sustainable Development Goal SDG 3.

The system collects information on users’ lifestyle habits, environmental exposures, and medical history through a questionnaire, then employs intelligent algorithms to conduct a comprehensive assessment of individual lung cancer risk. The core mechanism of the system is driven by an intelligent questionnaire generation and validation module. This module dynamically adjusts question content based on user responses to ensure the questionnaire’s logical relevance and consistency. Upon completion of the screening process, the system automatically generates a personalised risk-level report, accompanied by screening recommendations, thereby presenting health-risk distribution in a visualised format.

MAQ-SCREEN employs a multi-agent collaborative framework, thereby enabling synergistic operation among diverse agents through semantic reasoning and task-division mechanisms. It is evident that a range of agents are responsible for a variety of tasks, including the formulation of problems, the management of dialogue, the validation of answers, the analysis of data, and the generation of reports. The collaborative execution of these tasks by different agents serves to enhance the system’s accuracy and interpretability. This architecture is distinguished by its ability to not only calculate risks but also elucidate their origins, thereby addressing the lack of transparency that is inherent in conventional black-box models.

Additionally, the system incorporates a voice interaction module supporting both voice input and output, a feature particularly suited to users aged 45 and above who are

unfamiliar with keyboard operation. In order to ensure the protection of user privacy and data security, MAQ-SCREEN incorporates a data-protection module that employs encryption, access controls, and anonymisation mechanisms throughout the entire process of data collection, storage, and report generation. In essence, the overarching objective of MAQ-SCREEN is to evolve into a cost-effective, interpretable, and scalable solution for the purpose of early-stage lung cancer screening. The objective of this initiative is to enhance screening rates in resource-constrained settings, while concomitantly reducing reliance on costly LDCT examinations.

To ensure systematic development and evaluation, the MAQ-SCREEN project will be implemented in three structured phases, focusing on system design, prototyping, and evaluation. In Phase 1, the overall architecture and logic of MAQ-SCREEN will be defined, including questionnaire structure, lung cancer risk factors, and the interaction mechanisms among the five core agents (Questionnaire Designer, Dialogue Manager, Answer Validator, Data Analyst, and Report Generator). Deliverables of this phase include the system architecture and module design.

In Phase 2, a lightweight web-based prototype will be developed using Python (Flask) and JavaScript, integrating ASR/TTS modules and secure database connectivity. The multi-agent workflow will be implemented through message-passing coordination and semantic reasoning scripts, followed by internal debugging and functional testing to ensure system stability.

Finally, in Phase 3, pilot testing will be conducted with 10–15 participants in a simulated community-clinic environment to evaluate usability, accuracy, and accessibility. Feedback and performance data will be collected for iterative optimisation. The refined prototype will serve as the foundation for future large-scale deployment and academic evaluation. This progressive roadmap ensures the feasibility, interpretability, and scalability of MAQ-SCREEN as a practical early screening solution for lung cancer.

6. SDG Alignment

This project is closely aligned with the United Nations SDG 3—Good Health and Well-being. One of its key targets is to reduce premature mortality from non-communicable diseases through prevention and early intervention by 2030. The MAQ-SCREEN system supports this objective by leveraging a questionnaire-driven, multi-agent mechanism and AI-based inference to enable early lung cancer risk identification and personalised health assessment. By offering a low-cost, non-invasive, and accessible screening method, the

system facilitates early detection within community and primary healthcare settings, thereby reducing late-stage diagnoses and preventable deaths. Moreover, the inclusion of ASR and TTS-based voice interaction lowers digital barriers for elderly and underserved populations, advancing inclusivity in health technology. Through this combination of intelligent automation and human-centred design, MAQ-SCREEN contributes to disease prevention, promotes public health awareness, and reinforces the goals of health equity and sustainable healthcare envisioned by SDG 3 [12].

7. Benefits / Impact / Significance of Project

The MAQ-SCREEN system proposed in this project aligns closely with United Nations Sustainable Development Goal 3, marking a significant advancement in the application of intelligent digital technology within national cancer screening programmes. This system provides a robust complement to conventional LDCT screening through its low-cost, non-invasive and scalable approach, thereby enhancing accessibility and equity in early-stage lung cancer screening across Malaysia.

The system has been demonstrated to deliver substantial benefits for both individual users and healthcare systems. MAQ-SCREEN is a system that enables users to proactively monitor their health status. It achieves this through intelligent questionnaires that allow for personalised risk assessment. It provides a viable pathway for community-level initial screening, particularly for populations in remote areas or with limited resources. This approach has the potential to reduce the workload on healthcare institutions and to allow scarce LDCT resources to be allocated more efficiently towards high-risk groups.

Furthermore, MAQ-SCREEN is dedicated to enhancing public health awareness and encouraging proactive early screening. In practice, a significant number of individuals elect to postpone seeking medical attention due to concerns regarding cost, anxiety, or a lack of familiarity with the disease. The system has been developed to facilitate more accessible and user-friendly screening processes through voice interaction and multilingual support. The system has been designed to motivate high-risk groups to undergo timely examinations. This mechanism has been demonstrated to enhance nationwide awareness of lung cancer prevention and treatment, while concurrently promoting the adoption of preventive healthcare principles.

It is evident that, in the long term, the anonymised screening data collected by MAQ-SCREEN is of considerable value in supporting epidemiological research and the formulation of public health policy. This facilitates data-driven health governance and

resource allocation. In conclusion, MAQ-SCREEN illustrates how intelligent, accessible and explainable AI systems have the potential to transform healthcare delivery at the community level, thereby supporting Malaysia’s pursuit of universal health coverage and sustainable development goals.

8. Uniqueness of Proposed Solution

The MAQ-SCREEN system introduces a novel framework for early lung cancer screening through its multi-agent architecture, intelligent reasoning, and human-centred interaction design. Built upon the MetaGPT framework, the system replaces static, single-algorithm workflows with a dynamic collaboration among five agents—question generation, dialogue interaction, answer validation, data analysis, and report generation. These agents communicate via message passing to enable adaptive decision-making and semantic reasoning, creating a self-adjusting screening process that enhances flexibility, robustness, and scalability.

Designed as a web-based, cross-platform, and lightweight solution, MAQ-SCREEN allows direct access via browsers on multiple devices without requiring installation. This increases accessibility, particularly in community and primary healthcare settings. Integrated ASR and TTS modules provide a voice-enabled experience tailored for middle-aged and elderly users, promoting inclusivity and engagement through natural, conversational interaction.

The system also implements end-to-end data protection, with encryption, anonymisation, and strict access controls ensuring compliance with medical data standards. Overall, MAQ-SCREEN establishes an innovative and practical paradigm that integrates intelligent reasoning, interactive accessibility, and robust data security for early lung cancer risk identification.

9. Expected Outcomes

The primary anticipated outcome of the MAQ-SCREEN system is the establishment of an intelligent, accessible, and low-cost platform for early-stage lung cancer risk assessment. This platform empowers users to undertake preliminary screening independently via interactive questionnaires and voice-based interactions. Following the conclusion of the project and its subsequent implementation, it is anticipated that the system will yield

the following results:

Comprehensive Intelligent Screening Database:

- Capable of collecting, organising, and managing large-scale questionnaire-based lung cancer risk screening data;
- Capable of securely storing users' historical screening records, supporting follow-up comparisons and trend analysis;
- Capable of facilitating data sharing for research and public health policy purposes under anonymised conditions.

Multi-agent collaboration and reasoning capabilities:

- Capable of executing collaborative tasks among multiple agents, including problem generation, validation, data analysis, and report generation;
- Capable of conducting real-time reasoning and self-verification to enhance screening accuracy and interpretability;
- Capable of producing explainable AI health assessment outcomes, enabling users to comprehend screening logic and risk rationale.

Accessibility and user experience:

- Capable of cross-platform web access without requiring application installation;
- Capable of providing voice interaction support (ASR and TTS modules) for elderly or digitally less proficient users;
- Capable of automatically generating personalised, visualised screening reports alongside corresponding health recommendations.

Data Security and Privacy Compliance:

- Capable of implementing encryption, anonymisation, and access controls throughout the entire data collection, storage, and transmission process;
- Capable of adhering to digital healthcare data security standards and ethical guidelines;
- Capable of enhancing user trust through transparent data management mechanisms.

10. Status of the project

New Project.

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Appendix

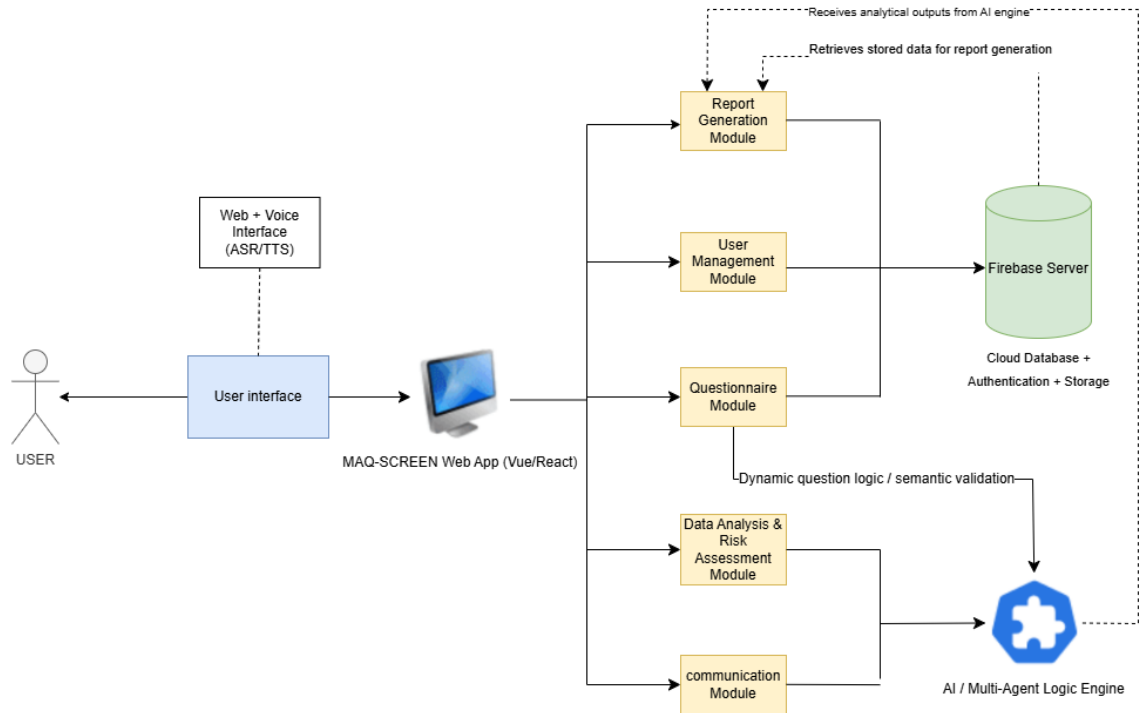


Figure 2: Architecture diagram of MAQ-SCREEN

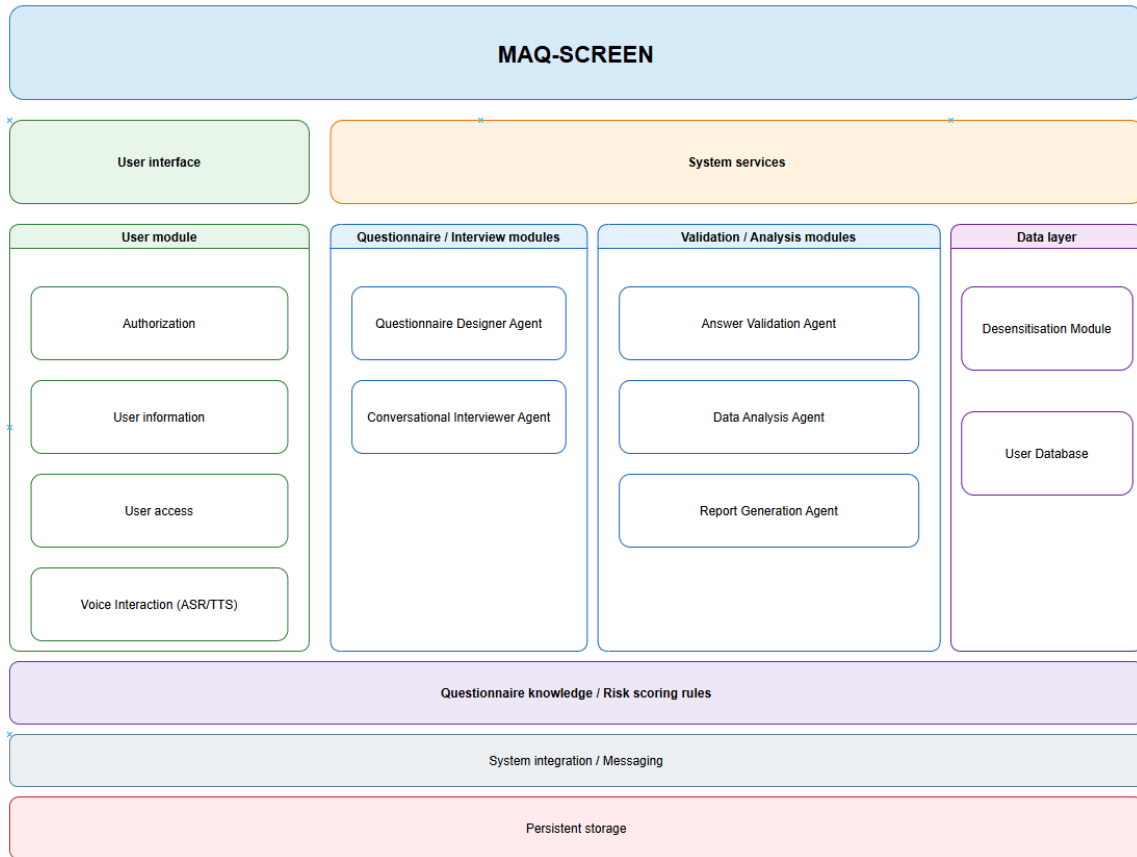


Figure 3: Module diagram of MAQ-SCREEN

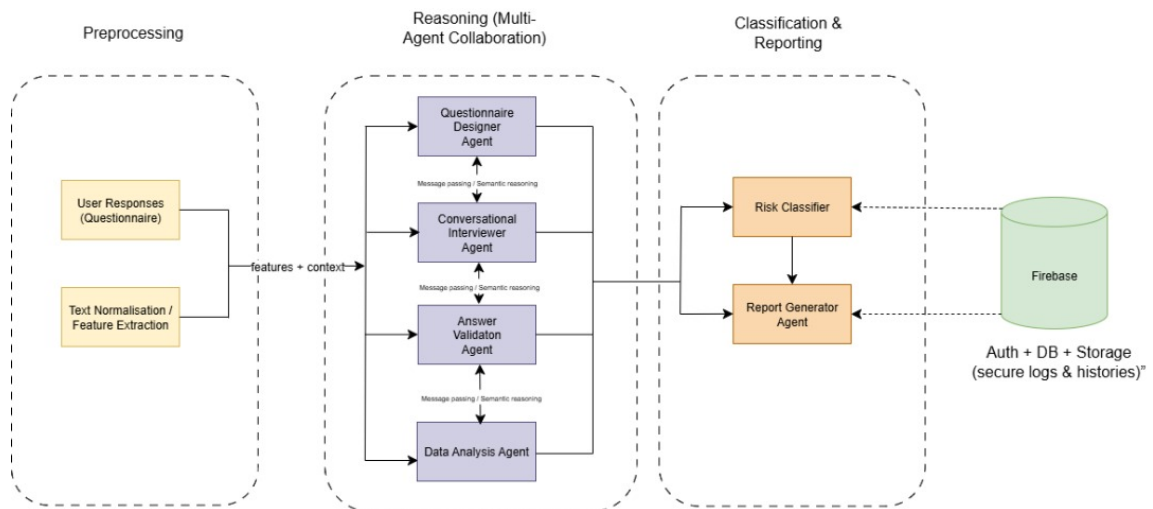


Figure 4: The flowchart of MAQ-SCREEN